

Statistics for Physical Scientists Short Course

Introduction

Welcome to this short course in statistics! It's designed to give researchers, particularly in the physical sciences, some tips, examples and practice in applying common statistical tools. The course covers:

- basic summary statistics, probability distributions and data combinations,
- overview of the Frequentist and Bayesian frameworks,
- correlation testing and significance, and sample comparisons,
- hypothesis tests and p -values,
- model-fitting and hypothesis testing using the χ^2 statistic,
- regression analyses (including least-squares and Gaussian processes),
- principal component analysis,
- practical error estimates (jack-knife, bootstrap and Monte Carlo simulations),
- propagating errors and Fisher matrix,
- Bayesian likelihood methods (including MCMC) and model selection.

Please note that this short course is intended to provide a broad overview of some practical statistics tools, appropriate for 20-30 hours' study. In your research you will likely be applying the statistical methods of your field in much more detail than is possible to cover here. However, I hope the overview will still be a useful starting point.

The course is structured in 6 classes, as described below, which are split into content presentation, worked examples and practical activities using the datasets provided. Each class comes with an accompanying **video presentation** and **python Jupyter notebook**, which provides summary notes and code for all the worked examples. I am an astrophysicist, so many of the activities tend to utilise astronomical datasets, even though the techniques are general.

At any time, you can approach the course in a self-study format and obtain support via e-mail or Zoom. However, once a year the course will run with **an additional 1-hour on-campus session each week** where we can discuss key concepts and you can obtain in-person support on the course activities. The next period of these in-person classes will be during **September-October 2024**. Please see the Canvas page for the scheduling details.

Please don't hesitate to contact me if you have any questions about the course: Chris Blake, cblake@swin.edu.au.

To complete the course ...

Please review the 6 classes (videos and/or slides) and complete the accompanying Jupyter notebooks. Once you have finished, please copy the final activity in each of the six classes

into a fresh (single) Jupyter notebook and e-mail that file to Chris Blake (cblake@swin.edu.au). These final activities are:

- Class 1: central limit theorem
- Class 2: analysis of neutron star masses (t-test, K-S test)
- Class 3: analysis of supernovae data (confidence regions for cosmological parameters)
- Class 4: Gaussian process fit to supernovae data
- Class 5: Monte Carlo simulation of Hubble's distance-redshift investigation
- Class 6: Akaike information criteria for curved/flat cosmological models

Useful books

The following is an (incomplete!) list of books which contain a great deal of practical wisdom in using statistics:

- Practical Statistics for Astronomers (Wall & Jenkins)
- Statistics for Nuclear and Particle Physicists (Lyons)
- Practical Bayesian Inference: A Primer for Physical Scientists (Bailer-Jones)
- Modern Statistical Methods for Astronomy (Feigelson & Babu)
- Principles of Data Analysis (Saha)
- Bayesian Logical Data Analysis for the Physical Sciences (Gregory)
- Data Analysis: A Bayesian Tutorial (Sivia)
- Numerical Recipes: The Art of Scientific Computing (Press, Teukolsky, Vetterling, Flannery)

Content outline

Class 1: Probability & statistics

- The process of science
- Common uses of statistics
- Summary statistics and their errors
- Estimators and bias
- Optimal combination of data
- Probability distributions: Binomial, Poisson, Gaussian
- The Poisson error
- Special role of the Gaussian distribution, central limit theorem
- Confidence regions and tails
- Comparing the Frequentist and Bayesian frameworks
- Monte Carlo simulations

Class 2: Correlation testing

- Correlation versus independence

- Pitfalls when searching for correlations
- Correlation coefficient
- Pearson product-moment correlation
- Significance of correlation
- Hypothesis tests and p -values
- Student t -distribution, tailed tests
- Non-parametric correlation tests; Spearman rank correlation
- Bayesian correlation methods
- Are the means of two samples consistent?
- Kolmogorov-Smirnov test

Class 3: Model fitting

- Comparing data and models
- The χ^2 statistic
- χ^2 probability distribution, degrees of freedom
- Reduced χ^2
- Use of χ^2 statistic as hypothesis test
- Modification for correlated data
- Use of χ^2 statistic for parameter fitting
- Errors in parameters from $\Delta\chi^2$; joint confidence regions

Class 4: Regression

- Introduction to regression
- Least-squares linear regression
- Quantifying the regression fit
- The case of errors in both co-ordinates
- Principal component analysis, eigenvalues and eigenvectors
- Interpolation
- Gaussian processes (kriging)

Class 5: Error estimates

- What is an error?
- Statistical versus systematic errors
- Error estimation using approximate sampling procedures
- Jack-knife errors
- Bootstrap errors
- Combining and propagating errors, linear and non-linear cases
- Fisher matrix forecasts
- Monte Carlo simulations

Class 6: Bayesian likelihood methods

- Bayesian Methods: conditional probability, Bayes' Theorem
- Role of the prior and likelihood
- Posteriors and confidence limits
- Marginalization
- Use of likelihood for parameter fitting
- Monte Carlo Markov Chains
- Model selection; is adding another parameter justified?
- Bayes factor and Jeffreys scale
- Akaike information criteria